

SECURE VLAN-BASED ENTERPRISE NETWORK WITH BRANCH CONNECTIVITY

SIMULATION AND IMPLEMENTATION USING CISCO PACKET TRACER

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01

PROBLEM STATEMENT

Modern organizations consist of multiple departments such as Employees, Students, Management, and Administration, all requiring reliable network access.

NETWORK SCENARIO:

If all devices operate on a single unsegmented network, several issues arise:

- High **broadcast traffic** leading to network congestion
- **Security risks** where users may access unauthorized resources
- Difficult **network management** as the network grows
- Lack of **reliability if servers fail**

PROPOSED SOLUTION:

To solve these challenges, a secure enterprise network is designed that:

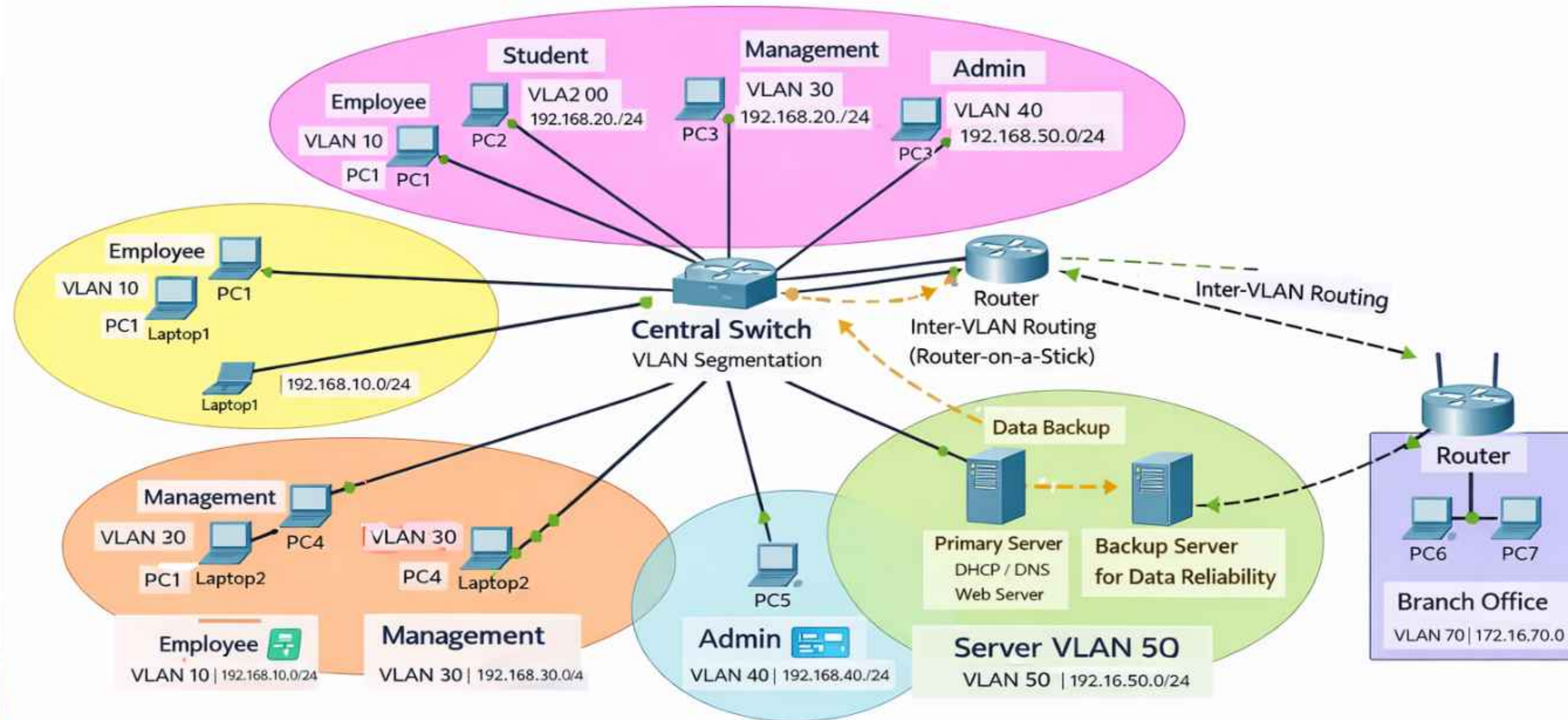
- Separates departments using **VLAN segmentation**
- Enables communication using **Inter-VLAN routing**
- Implements **Access Control Lists (ACL) for security**
- Provides services such as **DHCP, DNS, and Web Server**
- Includes a **Backup Server** to ensure service continuity and data protection.

NETWORK ARCHITECTURE

02

The designed enterprise network consists of the following components:

- A **central switch** connecting all departmental devices.
- A **router** performing **Inter-VLAN routing (Router-on-a-Stick)**
- Multiple **VLANs** representing different departments
- A **primary server** providing DHCP, DNS and Web services
- A **backup server** for data redundancy and service reliability
- A **branch office network** connected through a separate router



Key Design Features

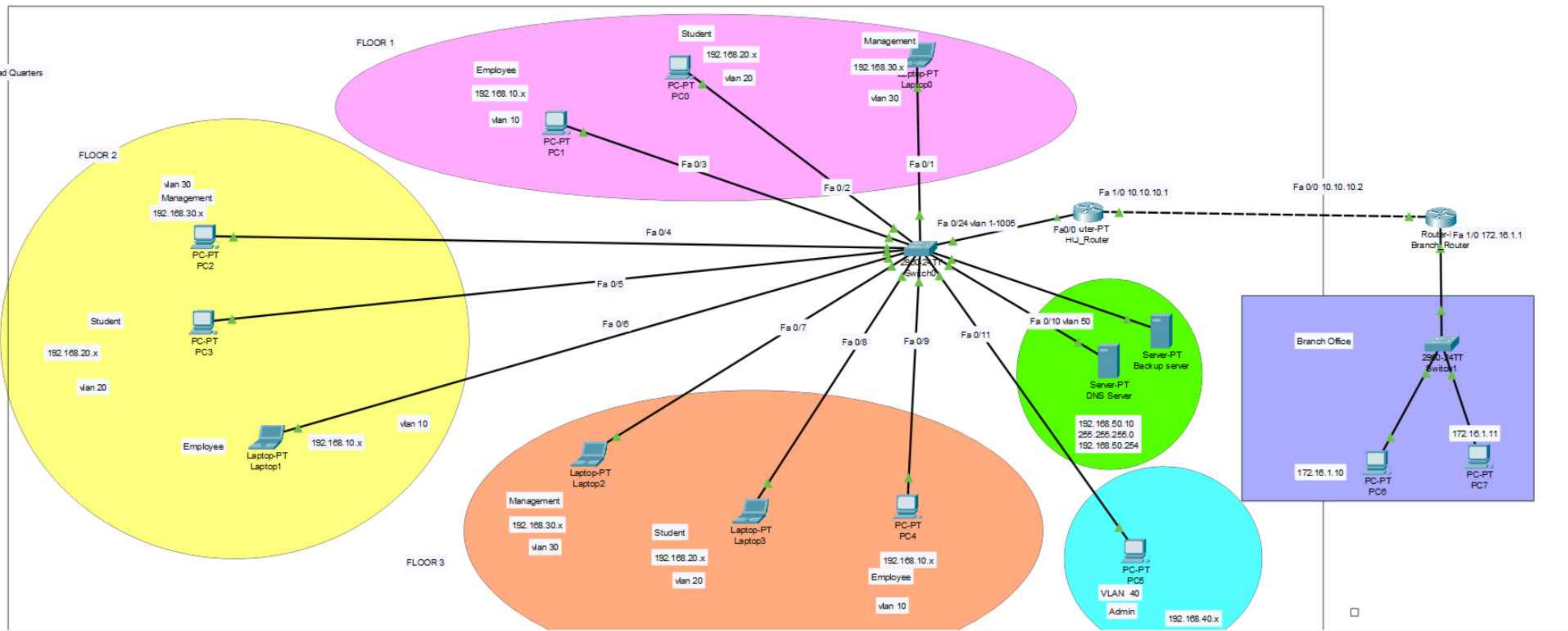
- ✓ Department-based VLAN Segmentation
- ✓ Secure Inter-Department Communication
- ✓ Centralized Network Services
- ✓ Branch Connectivity using Static Routing
- ✓ Backup Server for Fault Tolerance

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TOPOLOGY

cal x: 781, y: 452

Root 19:16



Realtime Simulation



Scenario 0

Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete

DEVICE SELECTION

01



ROUTER

- Performs Inter-VLAN routing
- Connects headquarters network to the branch office network

02



SWITCH

- Connects all network devices
- Implements VLAN segmentation

05



END DEVICES

- PCs and laptops used by different departments



PC-PT

04



Server-PT

BACKUP SERVER

- Stores backup copies of important server data
- Provides service continuity if the main server fails
- Improves network reliability

03



Server-PT

PRIMARY SERVER

- Provides DHCP for automatic IP allocation
- Provides DNS for domain name resolution
- Hosts the Web Server

VLAN DESIGN (SEGMENTATION)

05

To organize the network efficiently, each department is assigned a separate **Virtual Local Area Network (VLAN)**.

DEPARTMENT	VLAN ID	NETWORK ADDRESS
Employee	VLAN 10	192.168.10.0 /24
Student	VLAN 20	192.168.20.0 /24
Management	VLAN 30	192.168.30.0 /24
Admin	VLAN 40	192.168.40.0 /24
Server Network	VLAN 50	192.168.50.0 /24

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INTER-VLAN ROUTING

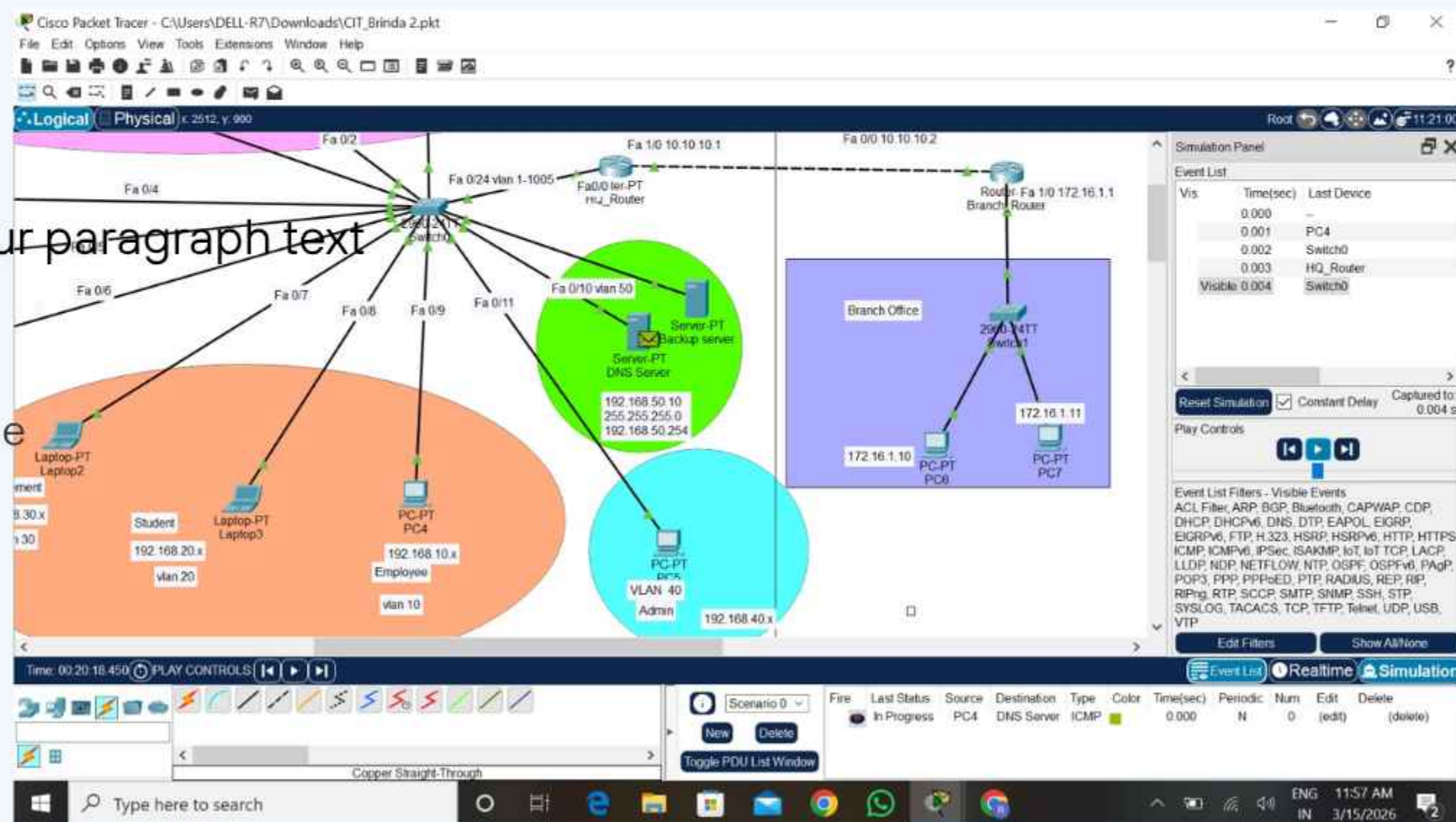
SIMULATION-
Employee pc to server
(Successful packet flow)

- The network is divided into multiple VLANs for different departments.
- Devices in different VLANs cannot communicate directly.
- A router is used to enable communication between VLANs.
- This is achieved using router-on-a-stick configuration with subinterfaces.
- Each VLAN has its own gateway IP configured on the router.
- This allows controlled communication between departments.

Gateway IPs

VLAN 10 – 192.168.10.254
VLAN 20 – 192.168.20.254
VLAN 50 – 192.168.50.254

Your paragraph text



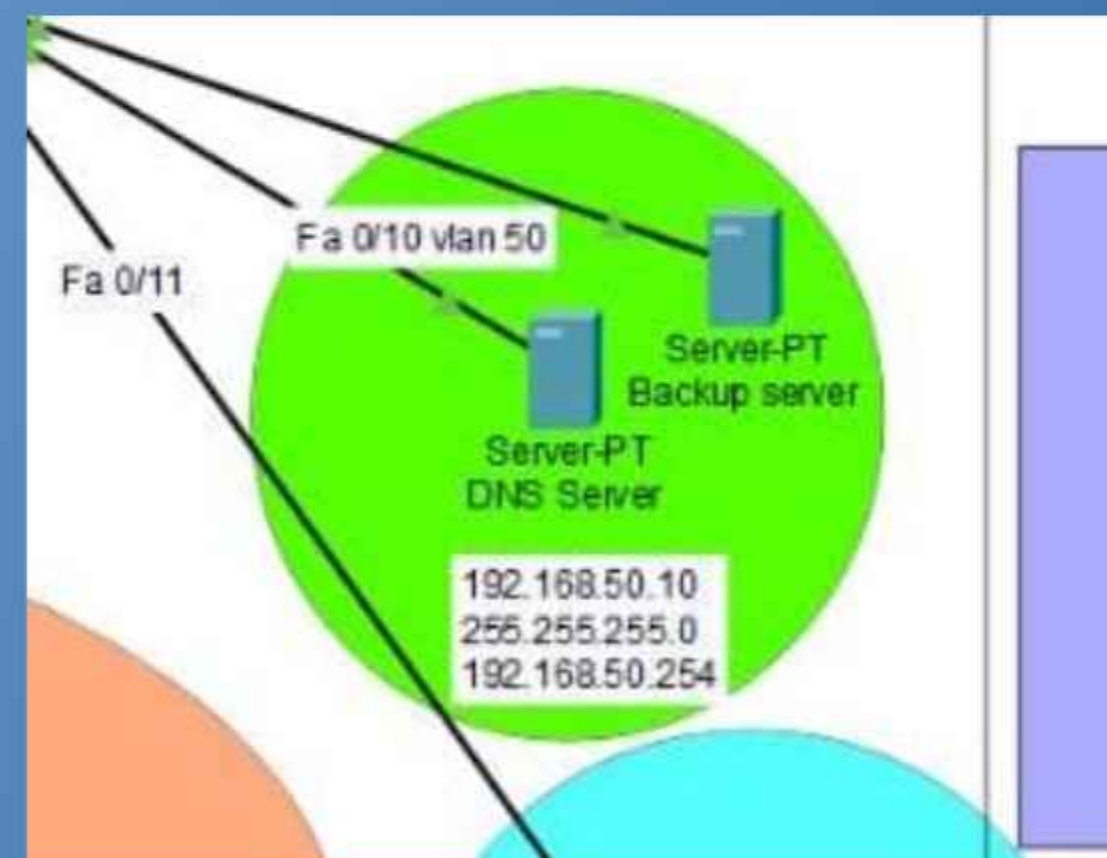
PRIMARY SERVER AND BACKUP SERVER

- A central server provides essential network services to all connected devices.
- The server supports DHCP, DNS, and web services for internal users.
- DHCP automatically assigns IP addresses to devices within the network.
- DNS allows users to access servers using domain names easily.
- A backup server is added within the same server VLAN network.
- It ensures continuous service availability if the primary server fails.

IP Address of servers

Primary Server – 192.168.50.10

Backup Server – 192.168.50.20



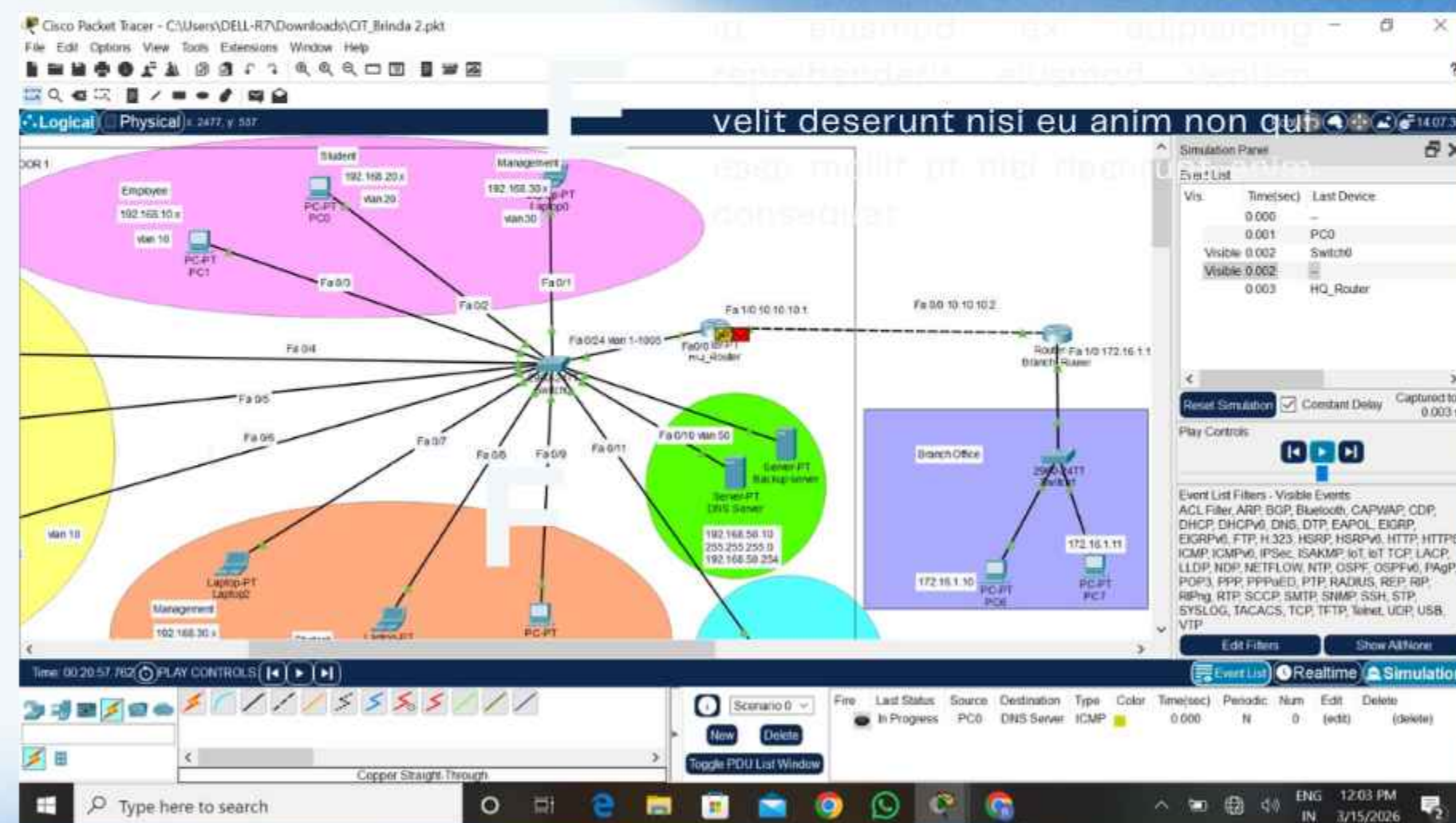
ACL AND NETWORK SECURITY

SIMULATION-
From student pc
(Packet blocked at router)

- Network security is implemented using an Access Control List (ACL).
- The ACL is configured on the router.
- It restricts unauthorized users from accessing the server.
- In this project, Person 2 network is blocked from accessing the server.
- Only authorized networks such as Head and Person 1 can access it.

ACL Rule

```
access-list 100 deny ip 192.168.20.0 0.0.0.255
host 192.168.50.10
access-list 100 permit ip any any
```

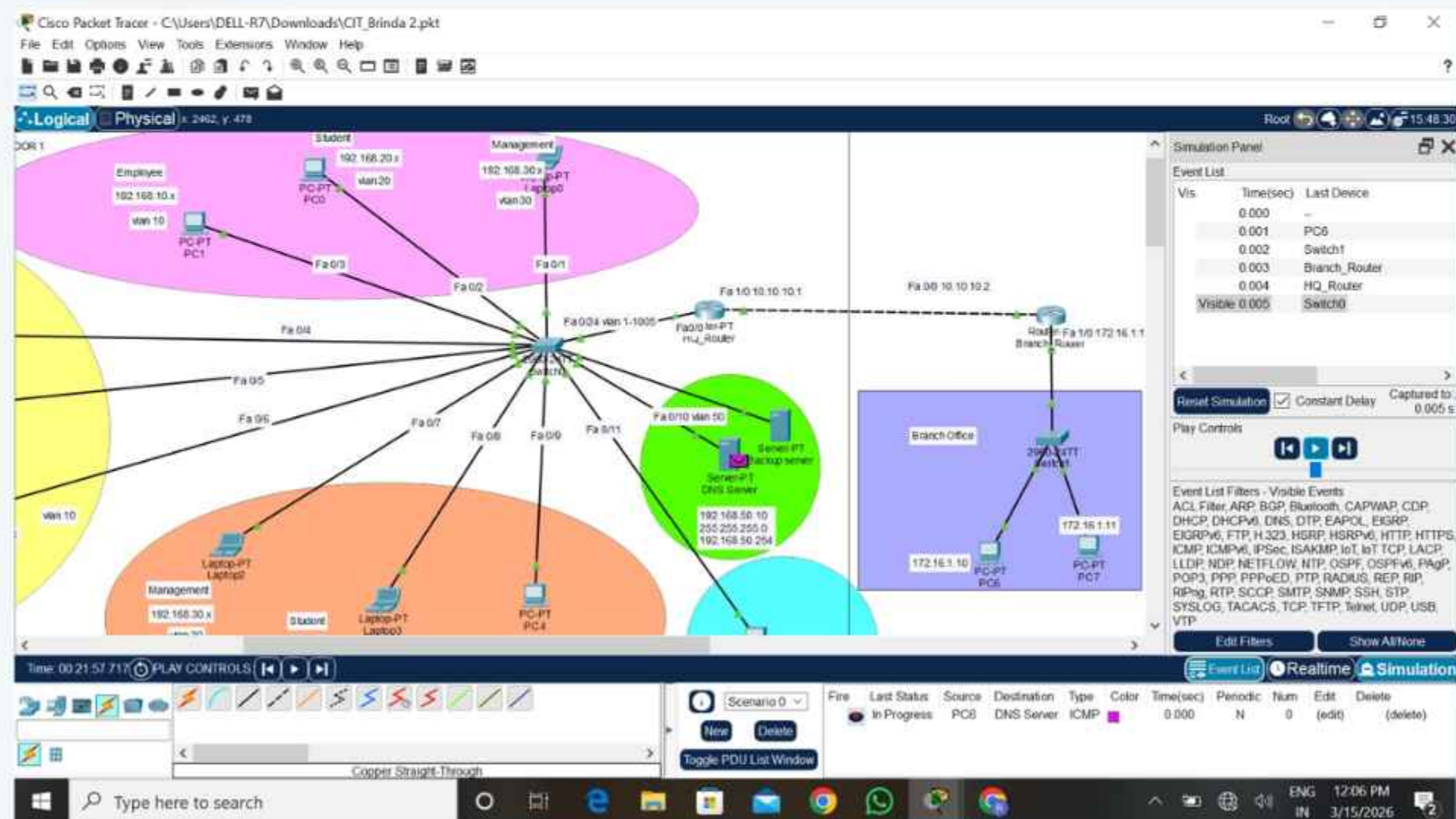


BRANCH NETWORK CONNECTION

- A branch office network is connected to the headquarters network.
- The connection is established using two routers.
- Static routing is used for communication between the networks.
- This allows branch devices to access central services like the server.
- It simulates a real organizational network structure.

Branch Network – 172.16.1.0 /24

SIMULATION–
Branch pc to server
communication



UNIQUE FEATURES AND CONCLUSION

- VLAN based segmentation organizes departments and improves overall network performance.
- Inter VLAN routing enables communication between different department networks securely.
- Access Control Lists restrict unauthorized users from accessing sensitive network resources.
- Centralized server provides DHCP DNS and web services to devices.
- Backup server ensures service continuity if the main server fails.
- Branch office connectivity demonstrates enterprise level network expansion and scalability.